2025-26 S1 – BIO220 PROGRAM

Week	Lecture	Tutorials and Labs	Coursework
Week 1	Introduction to computational Biology.		
Week 2	Charges and Coulomb law. Electrostatic potential.		
Week 3	Atoms and molecules. The Schrodinger equation		
Week 4	Molecular mechanics. Force Fields.		Coursework 1 (in class test): Molecular Physics.
Week 5	Structure of proteins and nucleic acids.	Tutorial and Lab: Molecular visualization with VMD.	
Week 6	Structural models and protein folding. Homology modeling and Alphafold.	Tutorial and Lab: Homology modeling.	Coursework 2 (laboratory): Homology modeling and molecular visualization.
Week 7	Molecular surfaces and electrostatics.		
Week 8	Molecular docking.		Coursework 2 deadline
Week 9	Energy minimization. Normal modes analysis.	Tutorial and Lab: Molecular docking, normal modes.	
Week 10	Free energy and solvation.		
Week 11	Small molecules and chemical spaces. Chemoinformatics.	Tutorial and lab: Protein/Ligand modeling.	Coursework 3 (laboratory): Protein/Ligand modeling.
Week 12	Drug libraries, screening and selection.		
Week 13	Recap.		Coursework 3 deadline

MOLECULAR MECHANICS AND FORCE FIELDS

Francesco Zonta, Ph.D.

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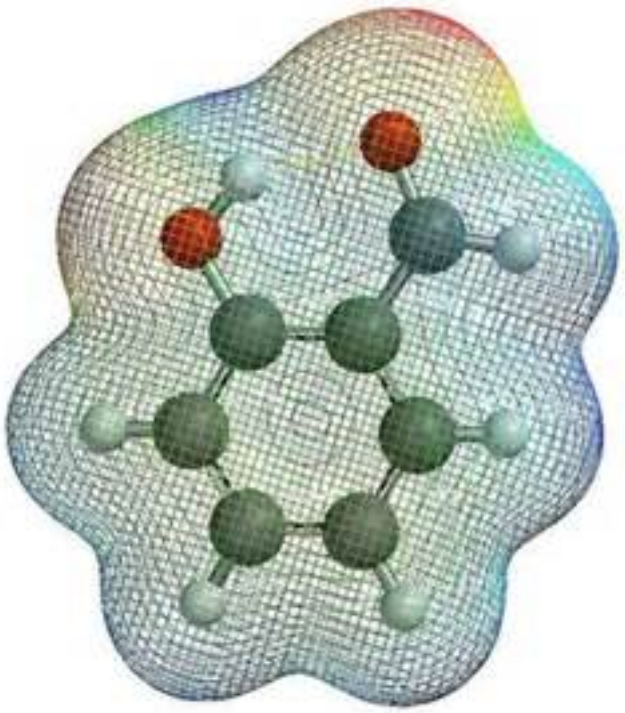
Room SB465, francesco.zonta@xjtlu.edu.cn



Xi'an Jiaotong-Liverpool University

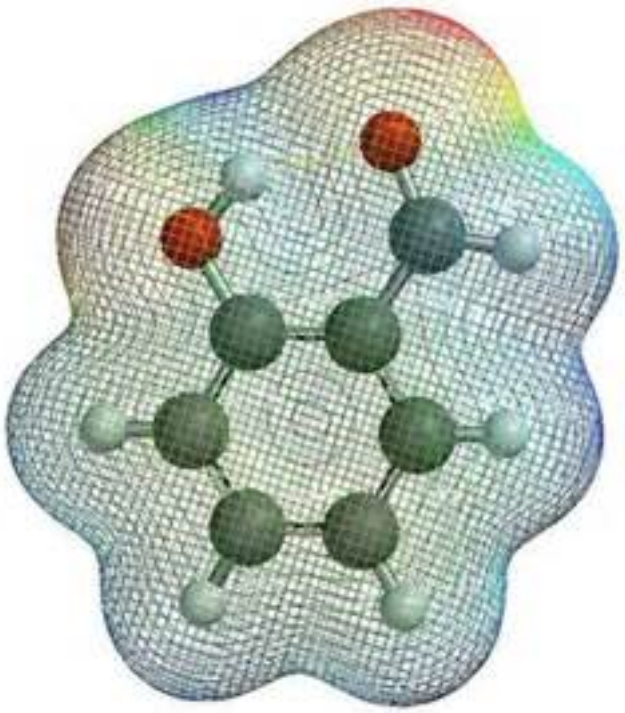
西交利物浦大學

Biology as seen by a Physicist



We want to describe our biological system of interest from the point of view of the Physics and **derive** its properties from physical principles.

Biology as seen by a Physicist



We want to describe our biological system of interest from the point of view of the Physics and **derive** its properties from physical principles.

From a theoretical point of view, a biological system can be regarded as a collection of nuclei and electrons, interacting through the electrostatic potential. The electron density can be written in term of a wave function, which is solution of the Schrödinger equation.

The Schrödinger equation – for multiple electrons

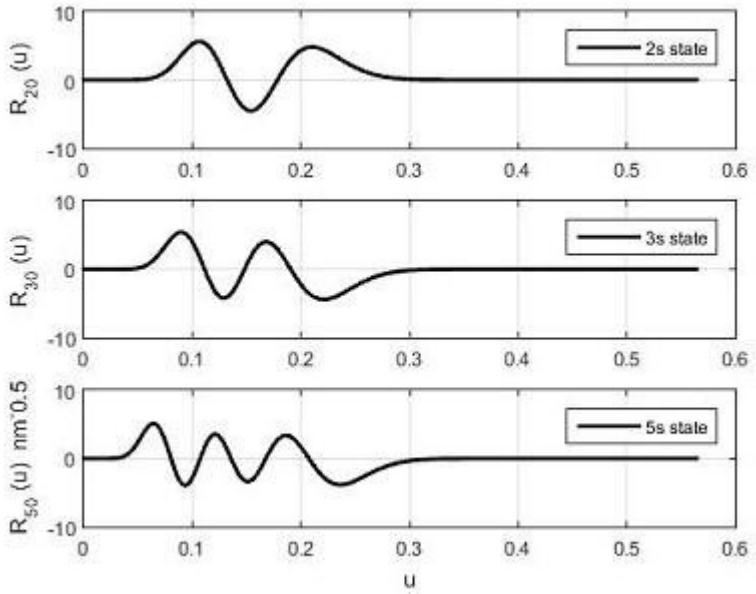


$$i\hbar\frac{\partial}{\partial t}\Psi(X,t) = \left[\frac{-\hbar^2}{2m}\nabla^2 + V(x,t)\right]\Psi(X,t)$$

$$\rho_i(X) = q_i|\Psi_i(X)|^2$$

charge density

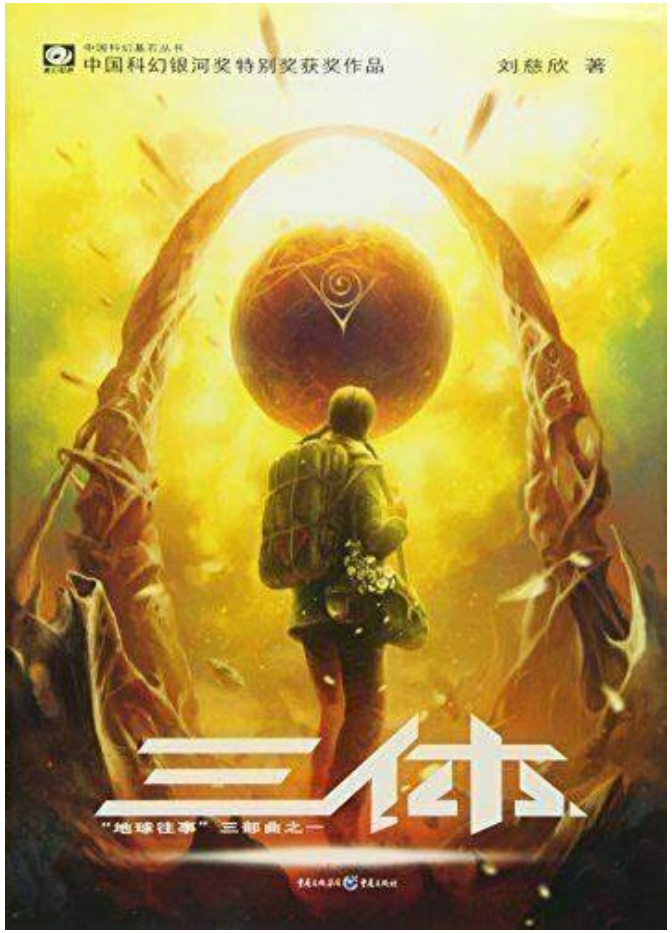
$$V_{1,2} = \iint \frac{\rho_1\rho_2}{4\pi\varepsilon_0 r_{1,2}} d\rho_1 d\rho_2$$



This equation describes the evolution of the probability density function in time.

At a fundamental level, atoms and electrons interacts only due to the electromagnetic field.

The Schrödinger equation – for multiple electrons



If you thought that the three body problem was difficult, it is only because you didn't hear about the quantum many body problem!

Solving the Schrödinger equation is a very complicated tasks even for small systems.

Systems with many interacting particles behave in a chaotic way that cannot be exactly predicted by physics principle (many body problems).

In order to solve the Schrödinger equation, approximations has to be introduced.

$$i\hbar\frac{\partial}{\partial t}\Psi(X,t) = \left[\frac{-\hbar^2}{2m}\nabla^2 + V(x,t)\right]\Psi(X,t)$$

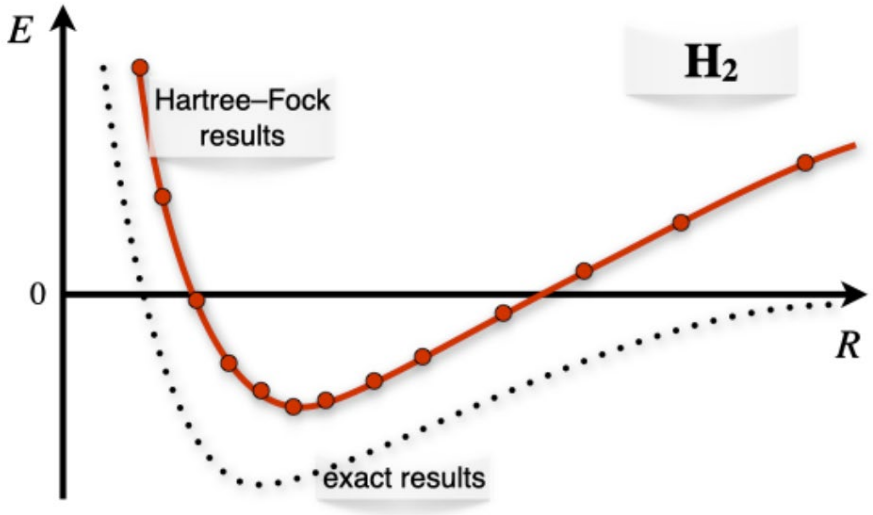
$$\rho_i(X) = q_i|\Psi_i(X)|^2$$

charge density

$$V_{1,2} = \iint \frac{\rho_1\rho_2}{4\pi\varepsilon_0 r_{1,2}} d\rho_1 d\rho_2$$

Hartree – Fock method





Hartree – Fock predictions for the chemical bond of H_2

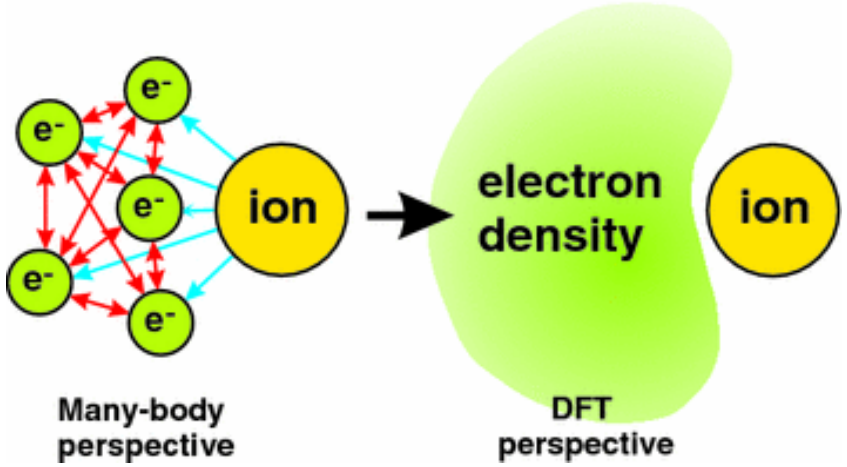
Solves the Schrödinger equation approximately by treating each electron as moving in the average electric field created by all other electrons (mean-field approximation).

Computationally expensive - scales roughly as N^4 with system size, limiting practical use to small molecules (~10-50 atoms depending on basis set).

Misses electron correlation effects - electrons avoid each other more than HF predicts, leading to systematic errors in energies and geometries.

Foundation for quantum chemistry - provides starting point for more accurate methods and useful qualitative molecular orbital pictures.

Density Functional Theory (DFT)



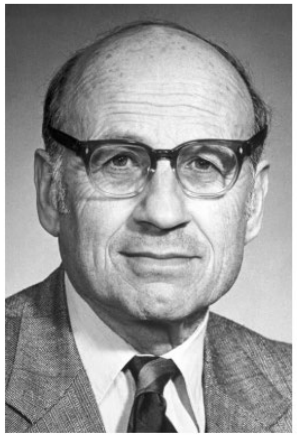


Photo from the Nobel Foundation archive.

Walter Kohn

Prize share: 1/2

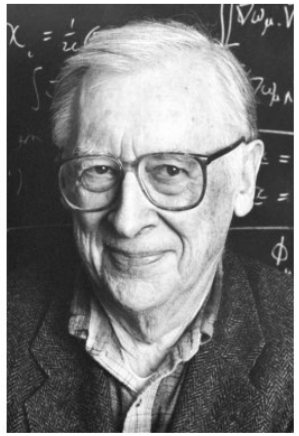


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John A. Pople

Prize share: 1/2

Nobel Prize in Chemistry 1998

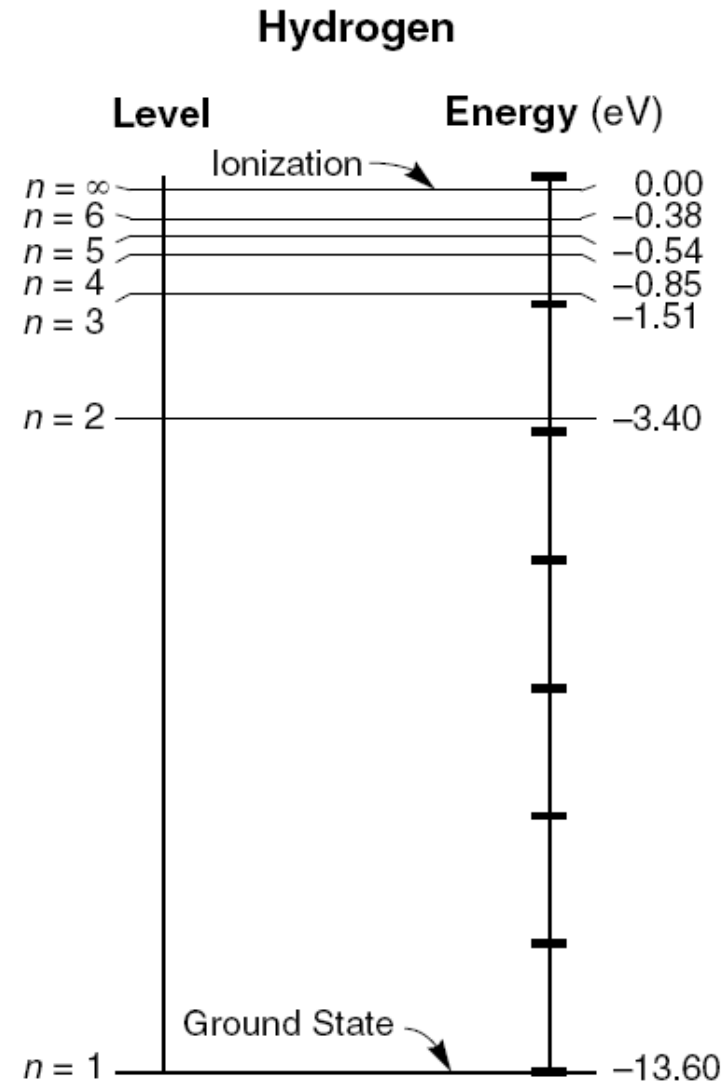
Uses electron density instead of wavefunction - reduces the complex many-electron problem to a simpler form while capturing most correlation effects.

Best accuracy-to-cost ratio - can handle systems of 100-1000 atoms, making it practical for small proteins, drug molecules, and active sites.

Widely used for parametrizing force fields - DFT calculations provide reference data (geometries, charges, energies) used to develop the classical potentials you'll learn next.

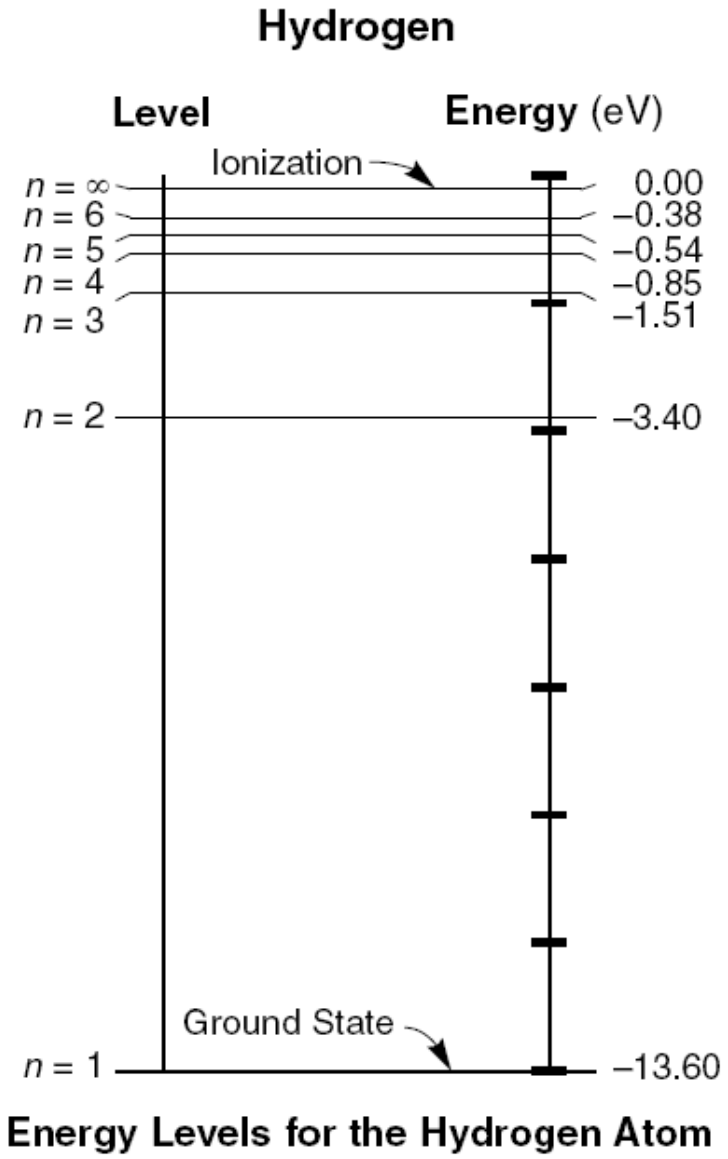
Still too expensive for macromolecules.


SO WHAT NOW?



Energy Levels for the Hydrogen Atom

Classical Physics still works!



Energy Levels for the Hydrogen Atom

Fortunately, we live at 310K, and biology uses only few *light* atoms

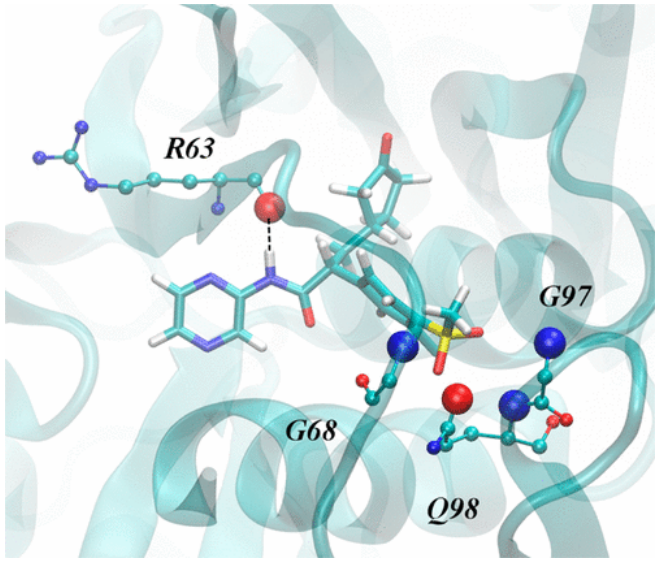
Thermal energy at room temperature: **26 meV**
Electronic energy levels: around **10 eV**
Probability of transition for thermal fluctuations: **10^{-171}**

The energy cost to excite light atoms is much higher than the thermal energy.

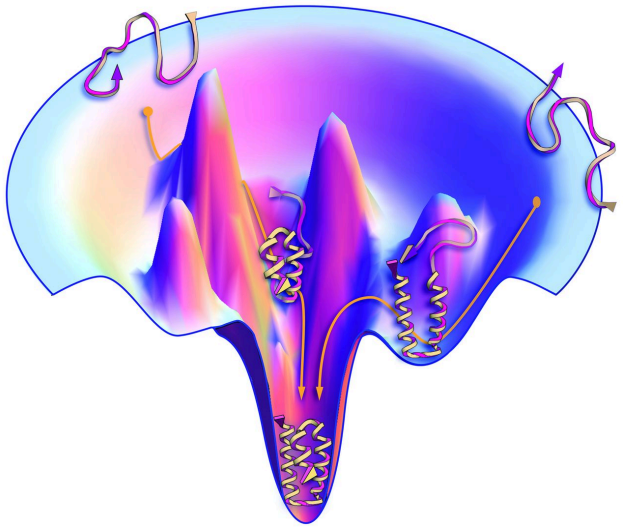
Light atoms (and molecules made of them) are all in the "ground state" at room temperature.

Chemical reactions and heavy atoms are very well controlled within any living organism.

Molecules according to classical physics



Mollica et al. J. Med. Chem. 2016



Each atom is treated as a “classical” object interacting with the other atoms in the system.

The interaction is represented through the classical potential energy.

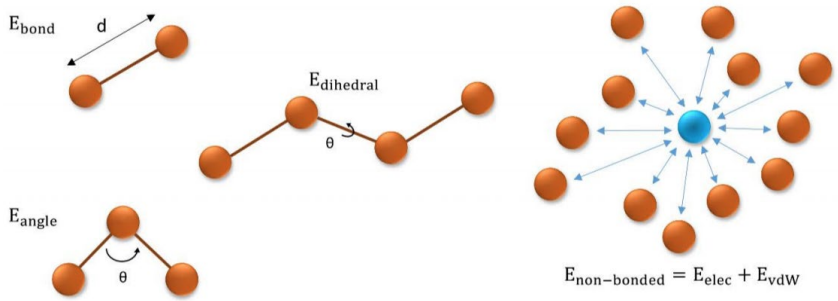
The equilibrium state is the minimum of the potential energy.

Different configurations are possible, without altering the electronic configurations of the molecules.

From the energy we can get the forces, and the time evolution.

At 310K different states are accessible, so Entropy exists, and Free Energy will decide the most probable state for the system.

Molecular Dynamics - The force is the gradient of the potential energy



Each of the particle in the system will feel a certain force, which is the gradient of the potential energy given by the interaction with all the other particles:

$$U_i = U_i(x, y, z)$$

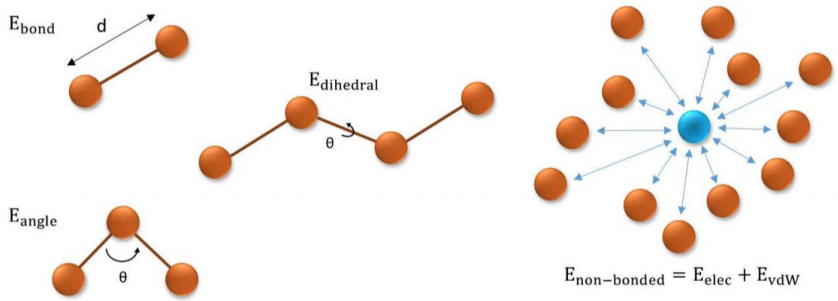
In this case, the force is given by the opposite of the *gradient* of the potential energy:

$$\boldsymbol{F} = -\boldsymbol{\nabla U_i} = \left(-\frac{\partial U_i}{\partial x}, -\frac{\partial U_i}{\partial y}, -\frac{\partial U_i}{\partial z}\right)$$

$$\boldsymbol{F} = \boldsymbol{ma}$$

Using the second Newton's law we can obtain the acceleration of each atom of the system if we know the force.

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$$\boldsymbol{F} = m\boldsymbol{a}$$

Using the second Newton's law we can obtain the acceleration of each atom of the system if we know the force.

BIO311

Molecular orbitals

Energy for excitation: 10 eV

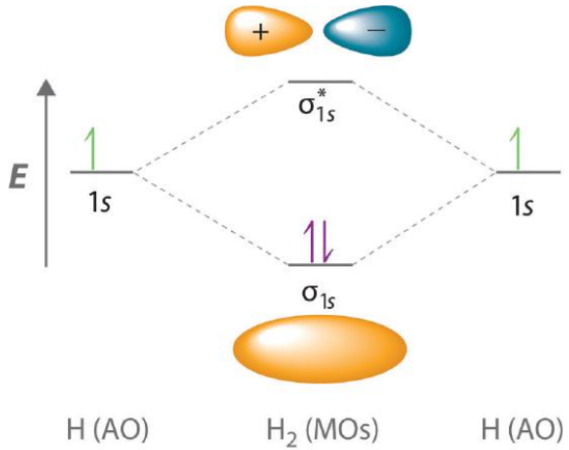
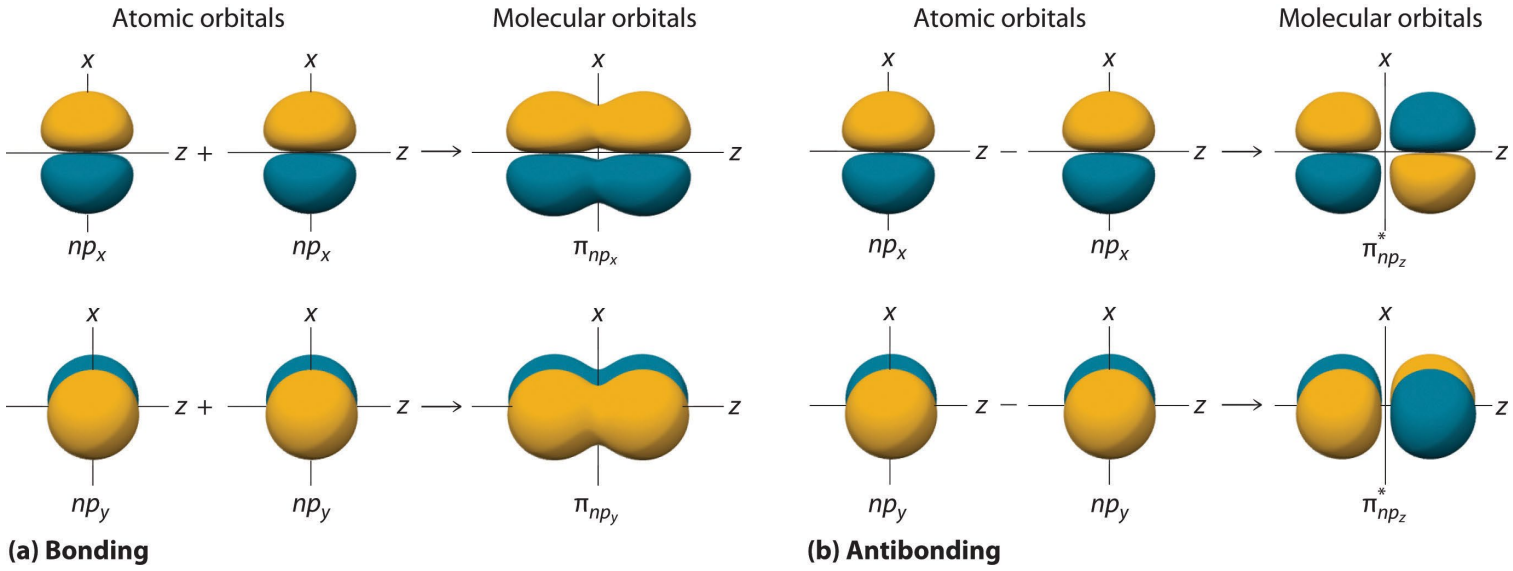
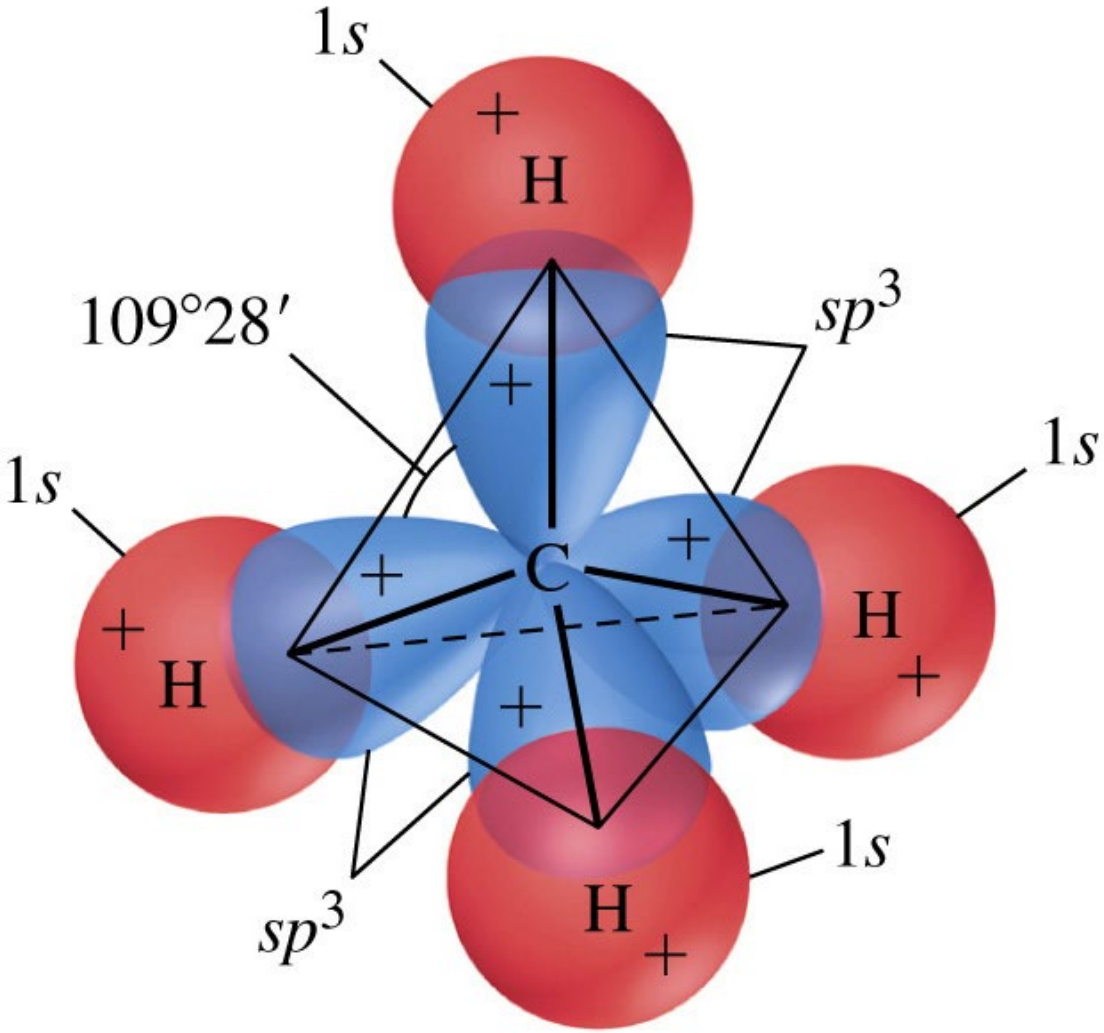


Figure 5.4.1.2: Molecular Orbital Energy-Level Diagram for H_2. The two available electrons (one from each H atom) in this diagram fill the bonding σ_{1s} molecular orbital. Because the energy of the σ_{1s} molecular orbital is lower than that of the two H 1*s* atomic orbitals, the H_2 molecule is more stable (at a lower energy) than the two isolated H atoms.



Classical Force Fields



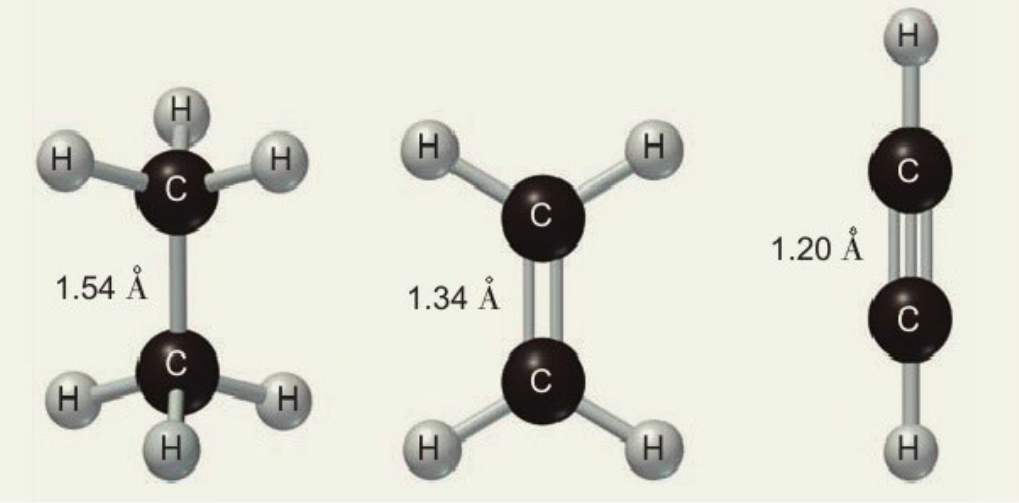
Since the electronic configuration is frozen in the ground state, we can simplify the form of the energy of interaction in many different ways:

1) A charge is assigned to each atom, and it is equal to the average charge of that atom.

2) Interactions between bonded atoms reproduce chemical and geometrical properties of the molecule, in particular regarding bond distances, angles and strength.

3) Non-bonded atoms interact exclusively through Coulomb and Lennard-Jones potentials.

Bond Distances



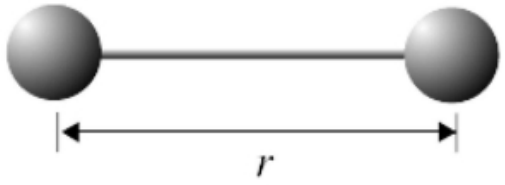
Bond type	k_r (KCal/mol Å^2)	r_0 (Å)
C–C	469.0	1.529
C=C	518.0	1.335
C≡C	865.0	1.177
C–N	427.0	1.475
C=N	570.0	1.297

The properties of the chemical bonds depend on the electronic configuration of the atoms in the molecule, and not only on the kind of atoms.

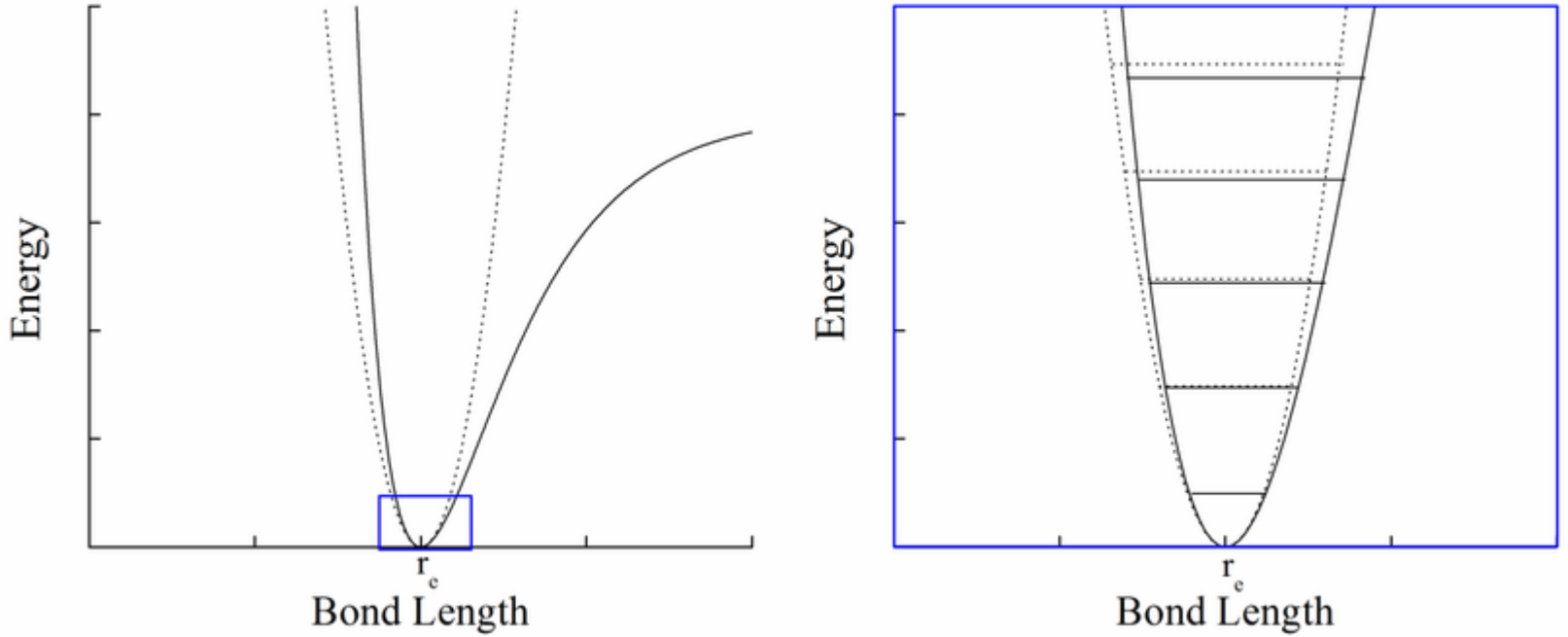
Atoms oscillate around their equilibrium average distances, thus chemical bonds are treated as *harmonic oscillators*.

The spring constant depend on the atoms involved and on their electronic configurations

$$U_r = k_r(r - r_0)^2$$



Harmonic potentials are approximations

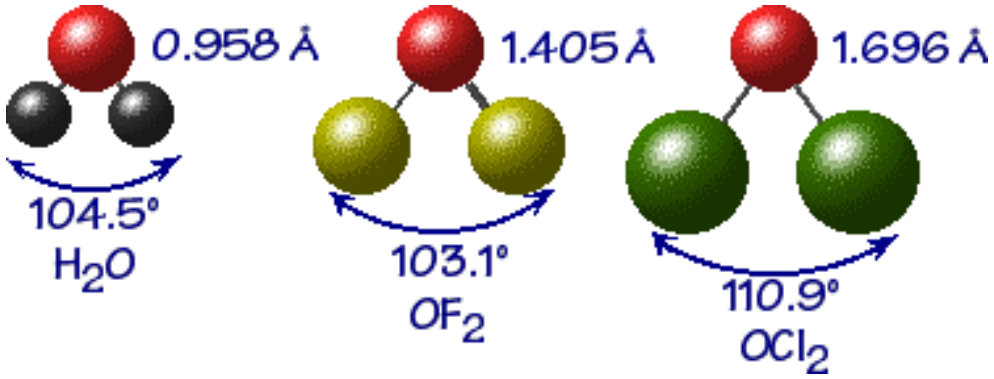


The harmonic potential is very rough approximation (not only because it is classical):

1) Distances between atoms cannot go below some certain threshold
2) When two atoms are very far, they do not interact.

It works only for very rigid bonds (higher level of energy are not excited – the difference is not noticeable).

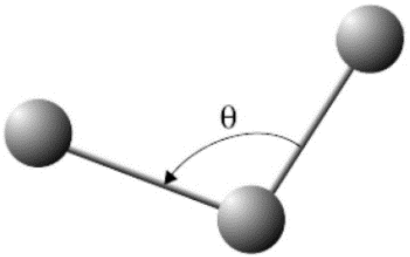
Bond Angles



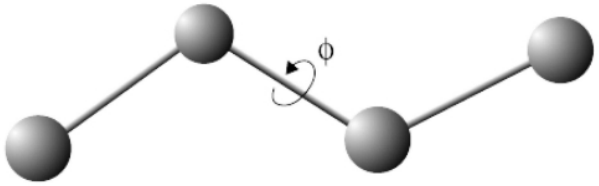
Two consecutive bonds form an angle.
Also for this angle, we assume a harmonic approximation: the angle can oscillate around the equilibrium configuration with a well-defined angular strength

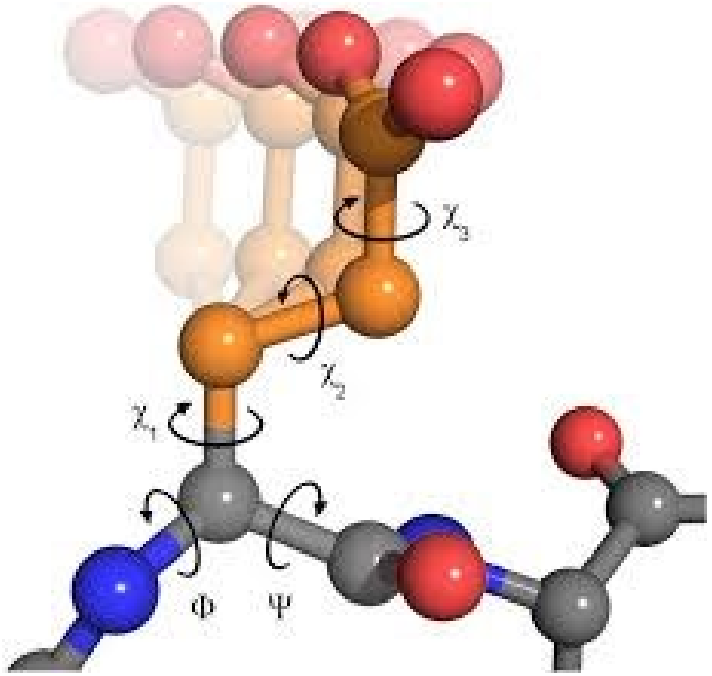
Angle type	k_θ (KCal/mol rad²)	θ_0 (°)
N–C–O	80	122.9
C–N–HN	33	118.0
N–C–N	80	116.6

$$U_\theta = k_\theta(\theta - \theta_0)^2$$



Dihedral Angles





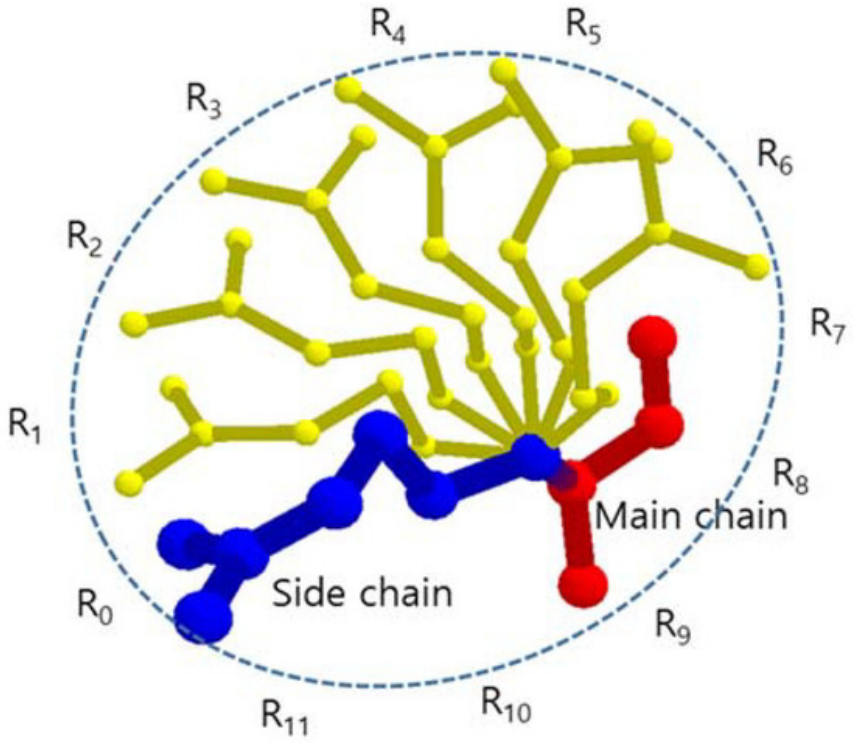
Four consecutive atoms define three bonds and two planes. The two planes describe an angle which is called *dihedral angle*.

Due to steric and electrostatic interaction, not all the angular position are allowed for the dihedrals. Several minima of the can exist for a single dihedral angle, corresponding to different possible configurations (*isomers of rotation* or *rotamers*).

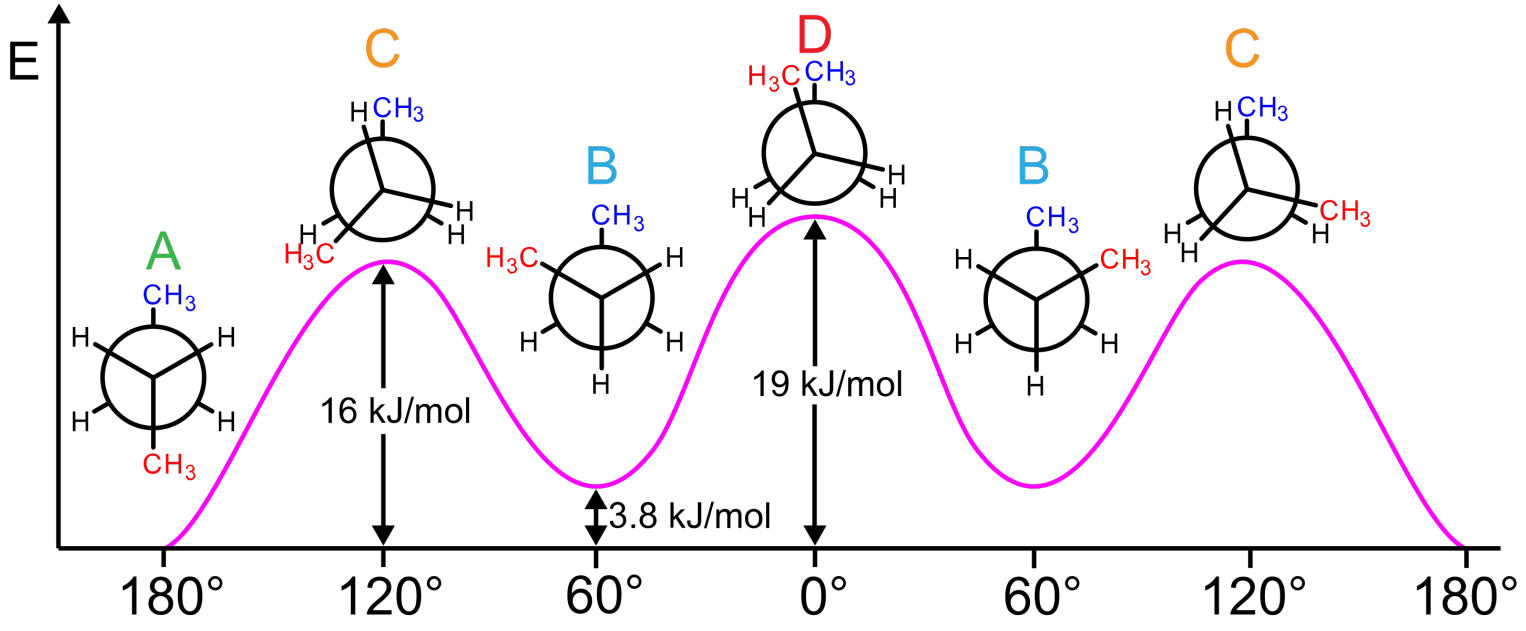
Dihedral angles allow biological macromolecules to be flexible and assume many different configurations.

$$U_\varphi = \sum C_k \cos(\varphi - \varphi_{0,k}))$$

Dihedral Angles

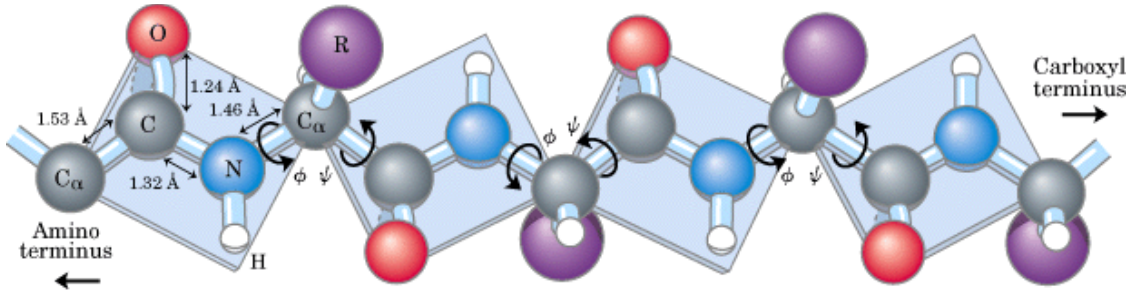


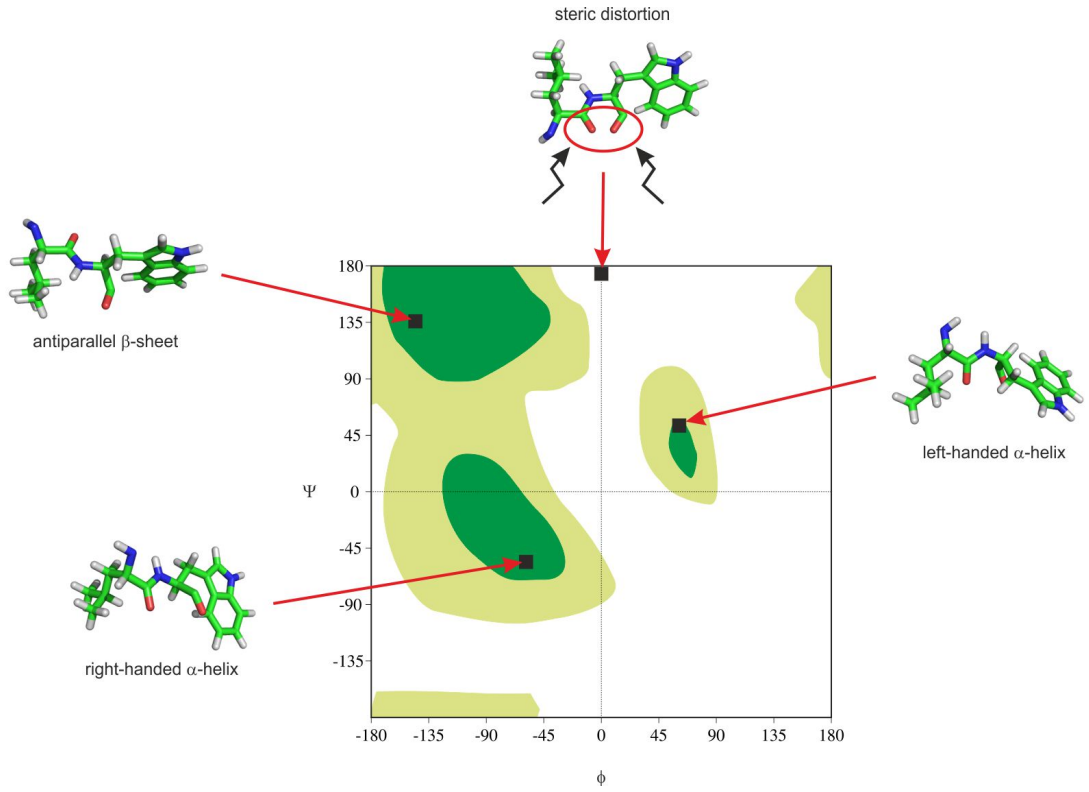
Seo et al. Molecules, 2018



Relative conformation energy diagram of butane as a function of dihedral angle

Ramachandran Plots



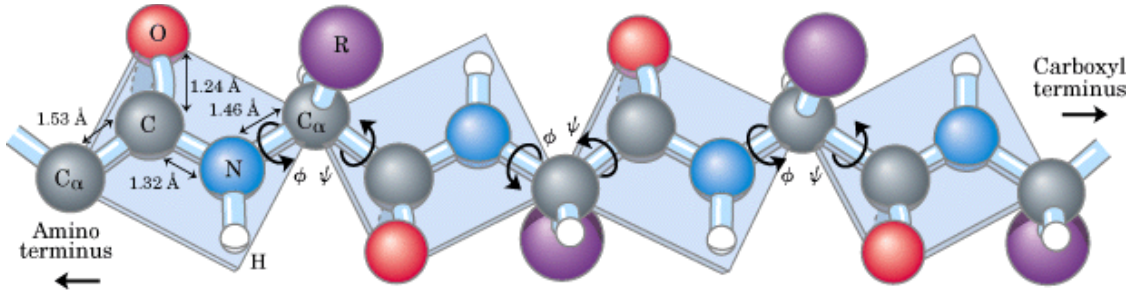


The relative position of two consecutive amino acids is described by two dihedral angles (φ and ψ).

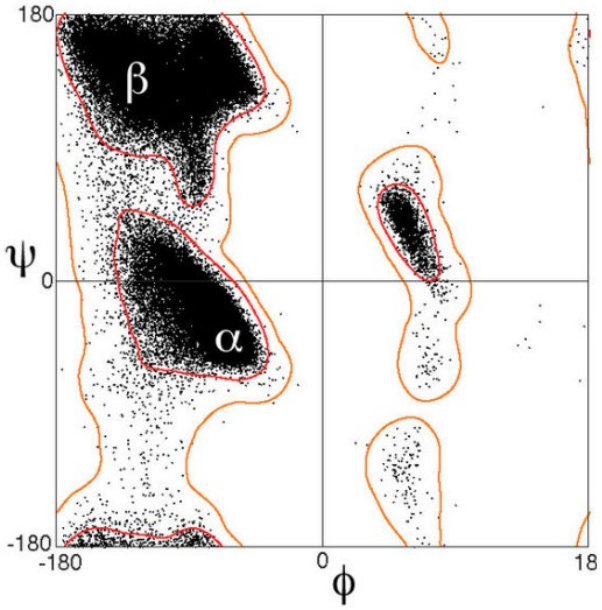
Not every combination of φ and ψ is possible, due to steric and electrostatic hindrance.

The values of the two dihedral angles is linked to the local secondary structure.

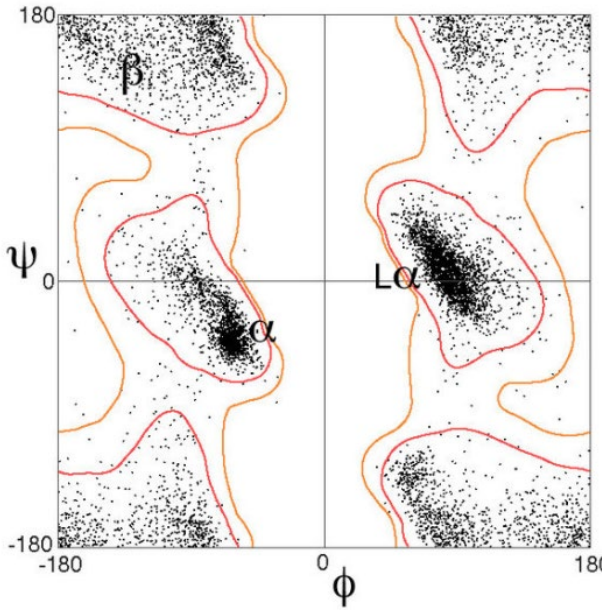
Ramachandran Plots



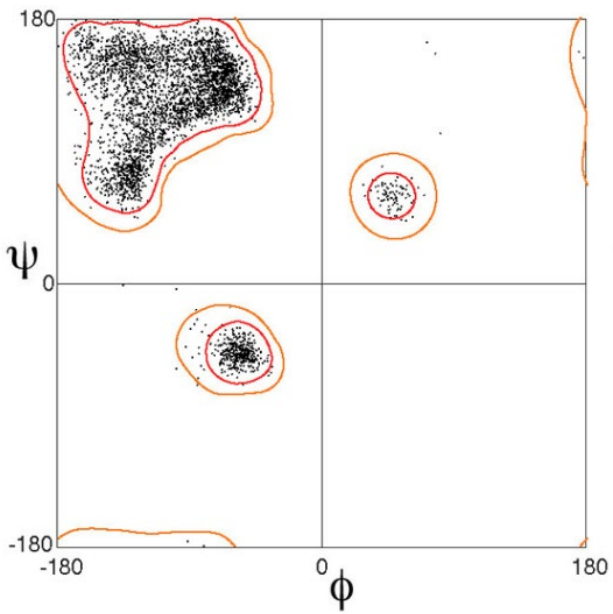
Different amino acids have different preferred orientation (which explain their secondary structure propensity).



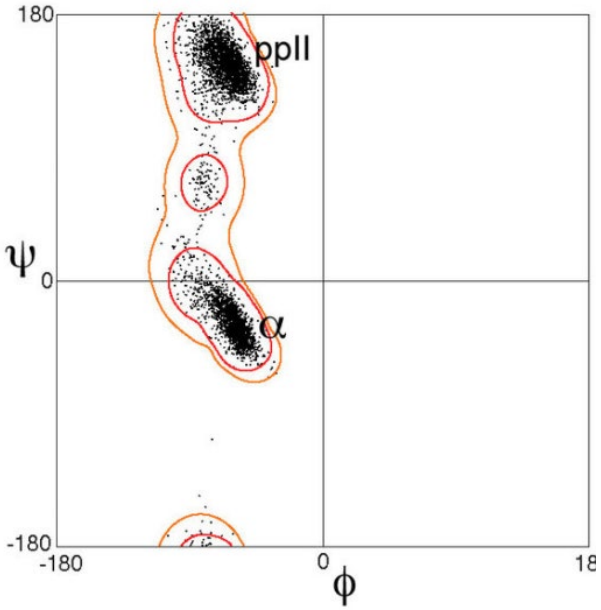
Average



Glycine



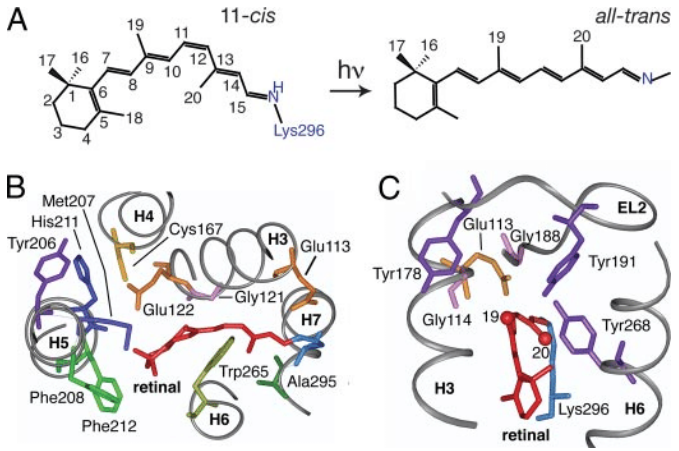
Proline



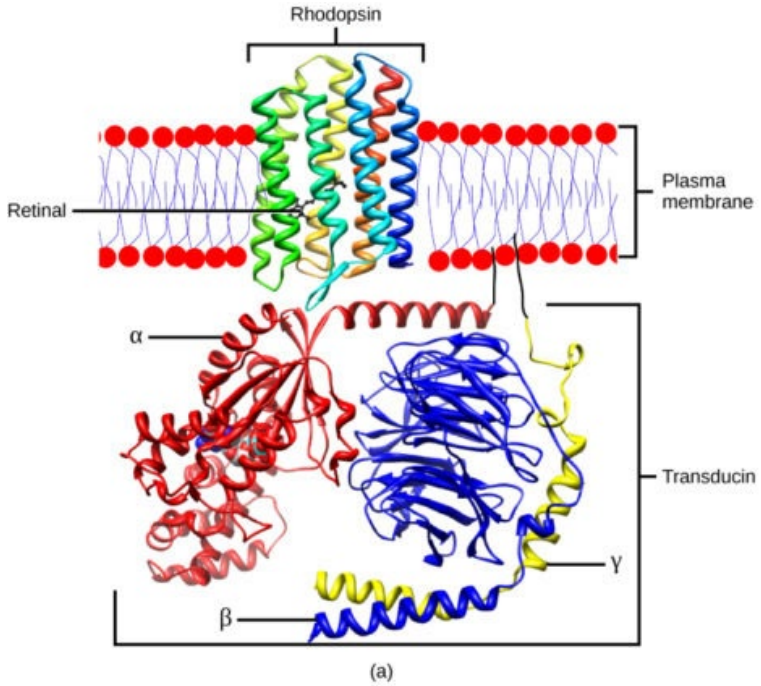
Pre-Proline

Francesco Zonta – Room SB465, Department of Biosciences and Bioinformatics – francesco.zonta@xjtlu.edu.cn

Cys-Trans isomerism



Ahuja et al. JBC, 2008



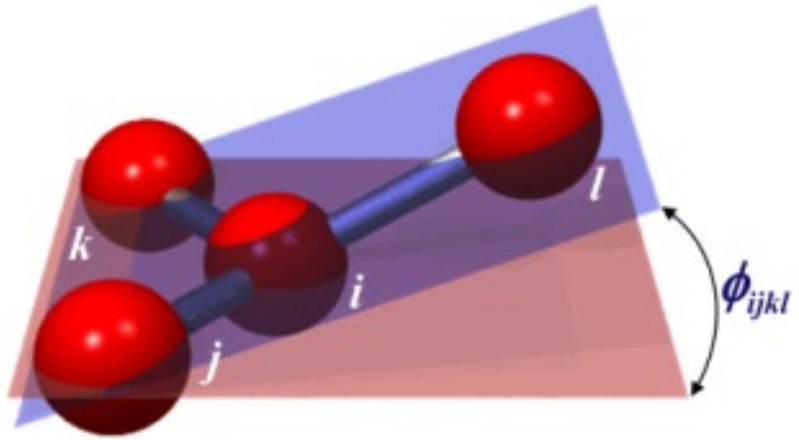
bio.libretexts.org

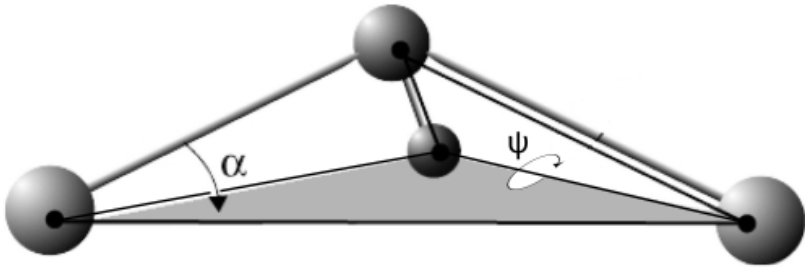
The *cys-trans* isomerism, is particular important in chemistry and biochemistry. It refers to cases in which a dihedral angle can have two different values. Switching from one to the other can induce large configurational changes.

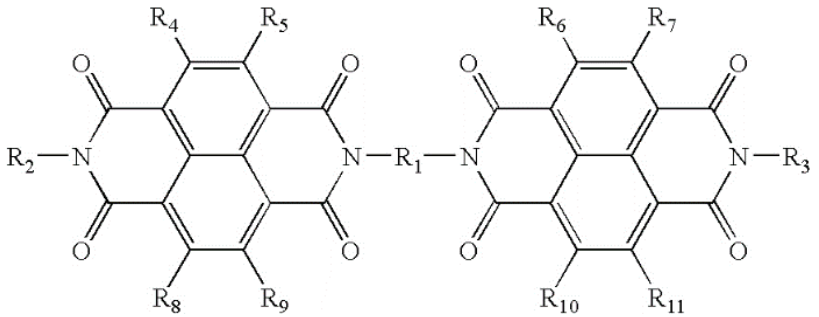
Rhodopsin is a subfamily of photosensitive GPCRs. It contains a *retinal*, a molecule that can transition between cys and trans configurations upon the absorption of a *photon.*

The change in the retinal configuration causes the activation of the GPCR signal, allowing the organism to sense light.

Improper dihedral angles







Carbon rings and other similar structure display planar configurations due to the specific electronic configurations of their atoms.

Improper dihedral angles need to be introduced to keep such atoms in the same plane.

$$U_\alpha = k_\alpha(\alpha - \alpha_0)^2$$

Non-bonded atoms

Atoms that are not connect by stable chemical bonds are considered non-bonded, and they interact through long range potentials:

Coulomb interaction: the charge represent the average charge on the atom due to the distribution of the electrons in that particular molecule.

$$U_C = \frac{q_i q_j}{4\pi\varepsilon_0 r}$$

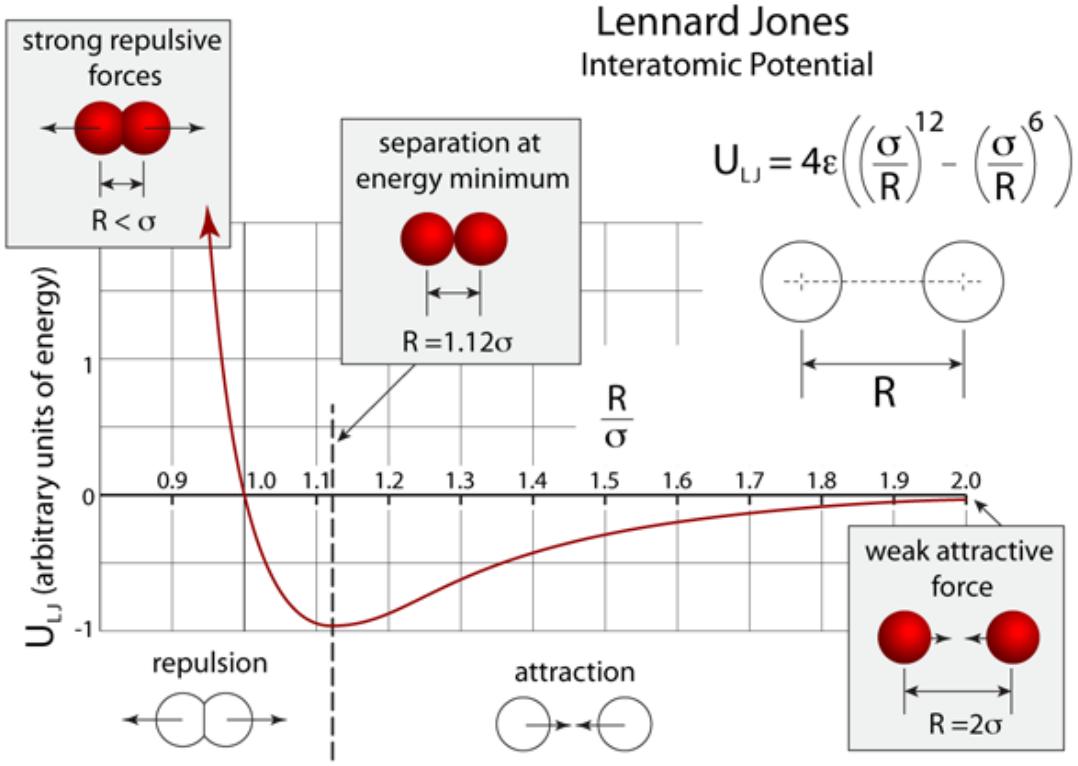
Lennard-Jones potential: It is the sum of an attractive part (quantum effect) and a repulsive part which represent the effect of electronic clouds in close proximity.

$$U_{LJ} = \left(\frac{A}{r^{12}} - \frac{B}{r^6}\right)$$

Hydrogen bonds: an empirical potential that take explicitly into account of the possibility of creation of hydrogen bonds.

$$U_{HB} = \left(\frac{C}{r^{12}} - \frac{D}{r^{10}}\right)$$

The Lennard-Jones potential



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At short distances, the LJ potential is strongly repulsive. Atoms that are too nearby in space create strong repulsions, that could be enough to completely distort the protein structure.

This happens very often when hydrogen atoms are added to the system (for example for a structure downloaded from the PDB or after a docking).

Energy has to be carefully minimized before starting MD.

Analytical form of the Potential Energy

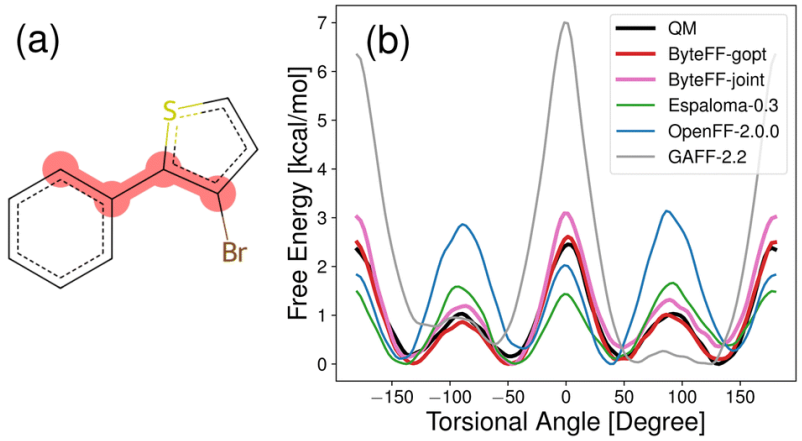
$$U = \sum_{bonds} k_r(r - r_0)^2 + \sum_{angles} k_\theta(\theta - \theta_0)^2 +$$

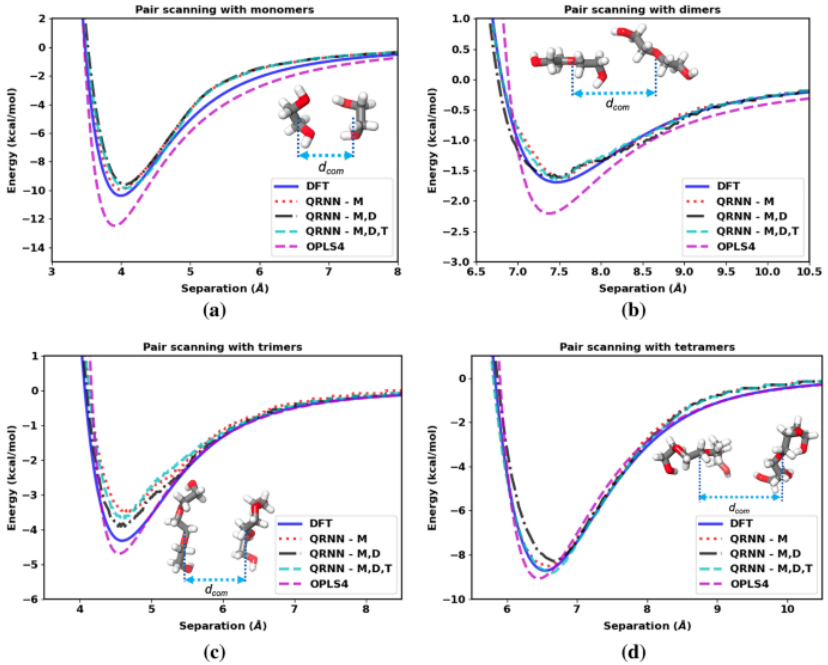
$$\sum_{dihedrals} C_k \cos(\varphi - \varphi_{0,k})) + \sum_{impropers} k_\alpha(\alpha - \alpha_0)^2 +$$

$$\sum_{i<j} \frac{q_i q_j}{4\pi\varepsilon_0 r_{ij}} + \sum_{i<j} \left(\frac{A}{r_{ij}^{12}} - \frac{B}{r_{ij}^{6}} \right) + \sum_{h.b.} \left(\frac{C}{r_{ij}^{12}} - \frac{D}{r_{ij}^{10}} \right)$$

This is one example of a classical force fields. Other possibilities exist.

How are parameters obtained in the energy function?





A Force Field is given by:

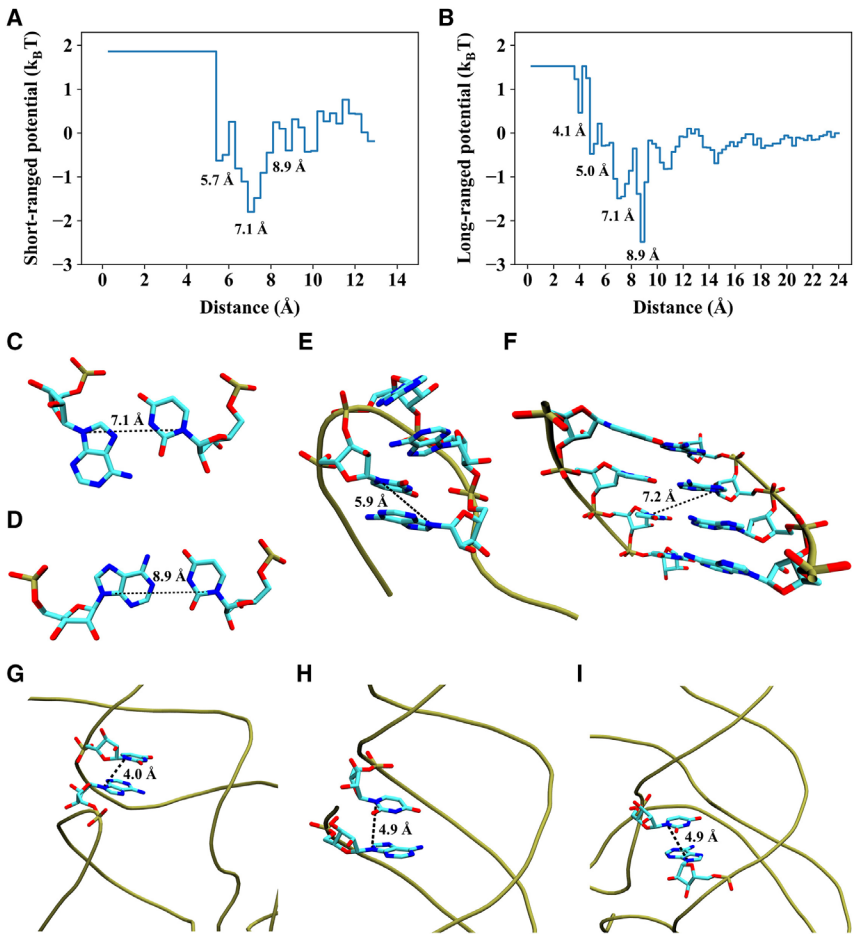
1) The functional form
2) The set of parameters

The functional form is decided because of theoretical reasons, and is the model used to represent the various molecules.

Parameters can be obtained either from experiments (molecular structures and properties) or from theory (quantum chemistry calculation, usually DFT).

In recent years, machine learning methods are emerging.
They typically learn the force fields from DFT results.

Statistical potentials



Derived from known structures - extracted from databases of experimentally determined protein structures (PDB) rather than from physical principles

Inverse Boltzmann relationship - assumes observed frequencies reflect underlying energies through the Boltzmann distribution:

$$P(\sigma) = e^{-E(\sigma)/kT}/Z$$

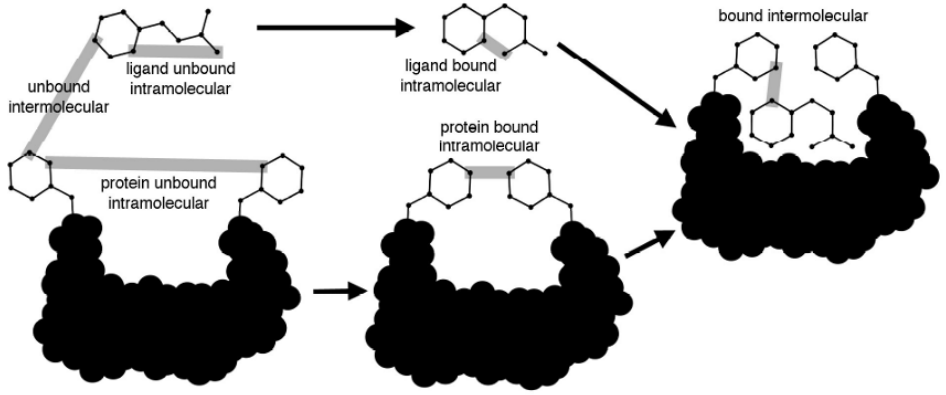
$$E(\sigma) = -kTln[P(\sigma) \cdot Z]$$

Practical implementation - count how often specific interactions occur (e.g., distance between residue types, contact frequencies, torsion angles) and convert frequencies into pseudo-energies

Advantages - computationally fast, implicitly capture complex effects difficult to model explicitly (desolvation, many-body interactions, entropic contributions)

Main applications - structure validation (is this model "protein-like"?), fold recognition, distinguishing near-native from misfolded structures

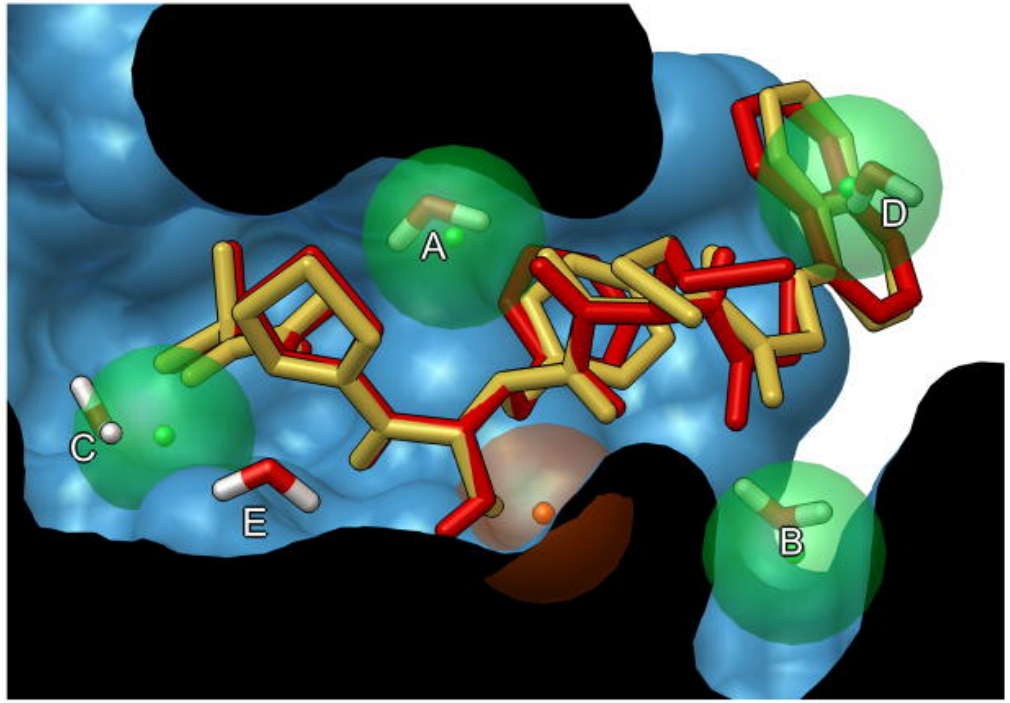
Molecular Docking: Autodock



The force field includes six pair-wise evaluations (V) and an estimate of the conformational entropy lost upon binding (ΔS_{conf}):

$$\Delta G = (V_{bound}^{L-L} - V_{unbound}^{L-L}) + (V_{bound}^{P-P} - V_{unbound}^{P-P}) + (V_{bound}^{P-L} - V_{unbound}^{P-L} + \Delta S_{conf})$$

where L refers to the "ligand" and P refers to the "protein" in a ligand-protein docking calculation.



Each of the pair-wise energetic terms includes evaluations for dispersion/repulsion, hydrogen bonding, electrostatics, and desolvation:

$$V = W_{vdw}\sum_{i,j}\left(\frac{A_{ij}}{r_{ij}^{12}} - \frac{B_{ij}}{r_{ij}^{6}}\right) + W_{hbond}\sum_{i,j}E(t)\left(\frac{C_{ij}}{r_{ij}^{12}} - \frac{D_{ij}}{r_{ij}^{10}}\right) + W_{elec}\sum_{i,j}\frac{q_i q_j}{e(r_{ij})r_{ij}} + W_{sol}\sum_{i,j}\left(S_i V_j + S_j V_i\right)e^{(-r_{ij}^2/2\sigma^2)}$$

Autodock potential is based on Physics, but it introduce weighting pre-factor to each term (W) to better fit empirical data.

How do we use scoring functions

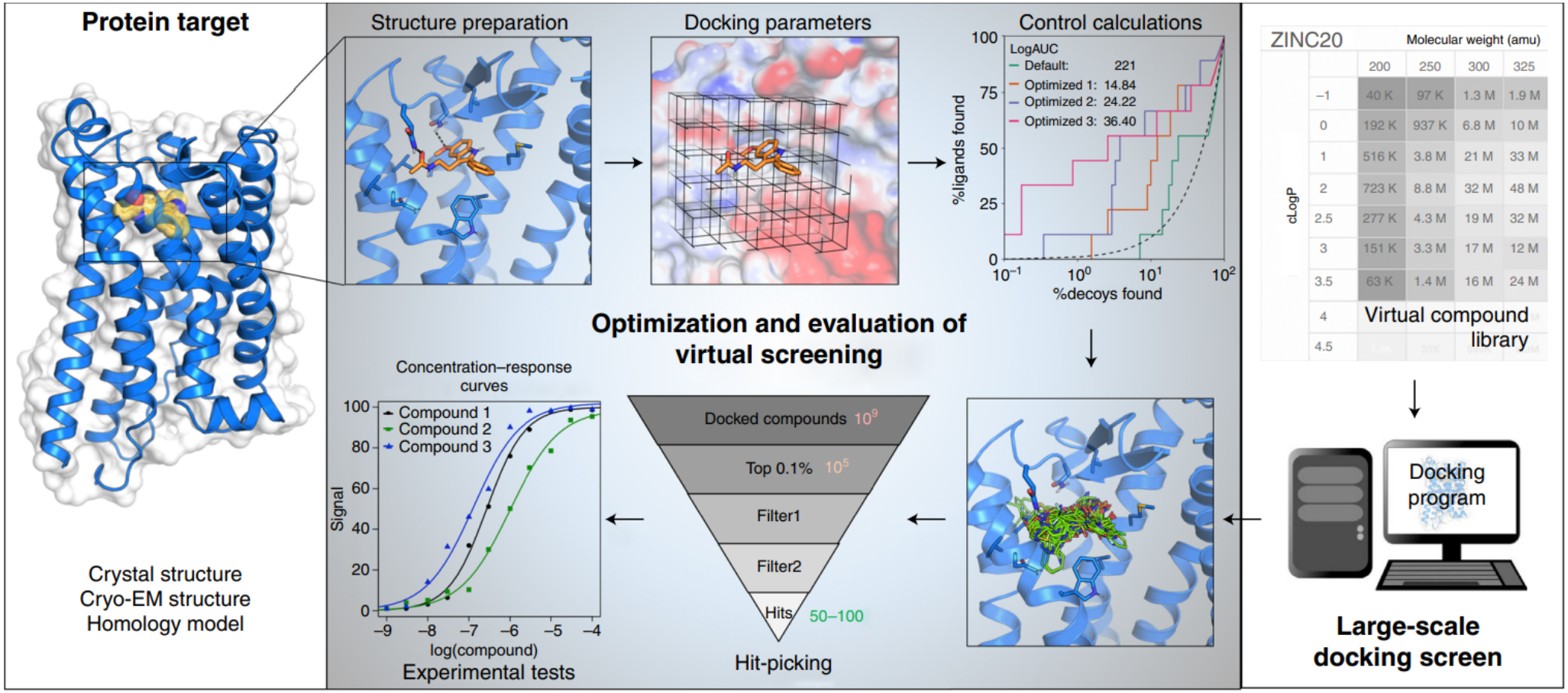


Fig. 1 | Large library docking workflow. The two required inputs for such a screen are the target structure and a screening database. Prior to using the database, the target structure must be converted into a representation used by the docking software and the pocket should be optimized with control calculations using retrospective analysis on known actives. After the prospective library has been docked, top-ranked hits can be filtered and selected for experiment. Multiple assays and controls are typically necessary to confirm activity.

Bender et. al. Nature Protocols 2021